

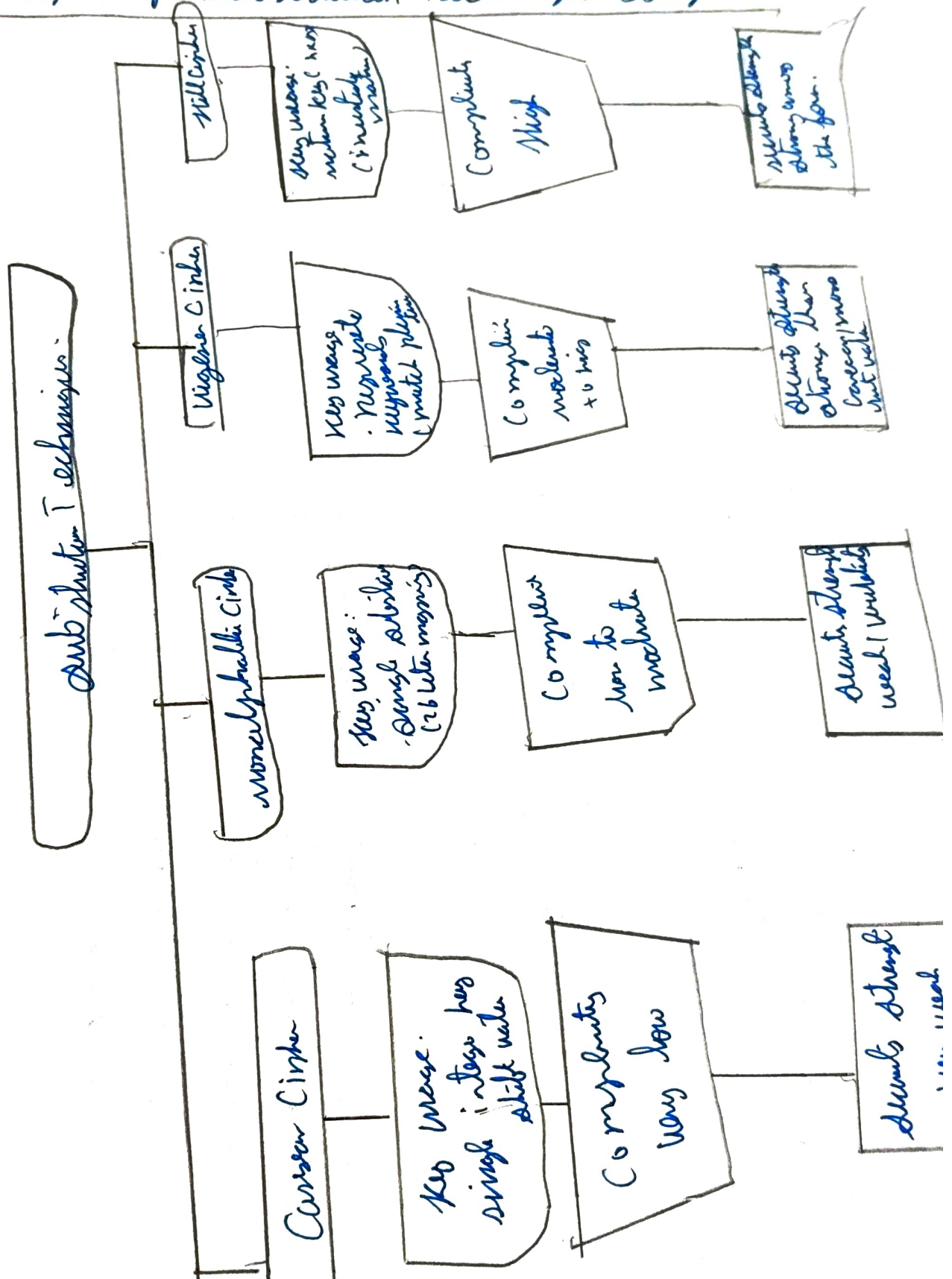
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Course: Cryptography and Network

Course code: C SA 5116

2. Concept map Substitution Techniques Comparison



3.

Concept map - Classical Encryption schemes.

Classical Encryption scheme.

Substitution Technique
Eg Caesar, monoalphabetic

- Change the character (value)
- Length and other remain the same

Transposition Technique
Eg Rail Fence, columnar permutation.

- Change the position (value)
- Character remain the same

Data Structure Impact

- Alphabet - change character value (same length - 26)
- Transposition - change order (same character)

Security level to measure

Adaptation to present analysis (vulnerable to present analysis)

Transposition slightly higher security

Advantages

- Both use cipher.
- Both protect confidentiality.
- Both can combine for better security

Limiting factor

- Substitution - monoalphabetic - Caesar, Vigenere, etc.
- Transposition - rail fence, columnar, etc.